

## Theorem 2.8 (The fundamental theorems of finite fields)

The size of every finite field is a power  $q = p^n$  of a prime number  $p$ , where  $n \in \mathbb{N}$ . Furthermore, for every prime power  $q$  there exists a finite field with  $q$  elements. This field with  $q$  elements is unique up to isomorphisms of fields, and it can be realized as  $\mathbb{F}_p[t]/g$  for some irreducible polynomial  $g \in \mathbb{F}_p[t]$  of degree  $n$ .

How would you find an irreducible of a given degree  $n$ ? For example, for the AES we need  $n=8$ .

A naive algorithm:

For a given polynomial  $g \in \mathbb{F}_p[t]$  of degree  $n$  one could test if it is irreducible by a brute-force

check. Just iterate over all polynomials

$f \in \mathbb{F}_p[t]$  with  $1 \leq \deg f < n$  and check

if  $g$  is divisible by  $f$ . If  $g$  is not divisible by any  $f$ , then  $g$  is irreducible.

Otherwise,  $g$  is reducible.

In this way, we do about  $p^n$  iterations.

This method can be optimized by replacing the

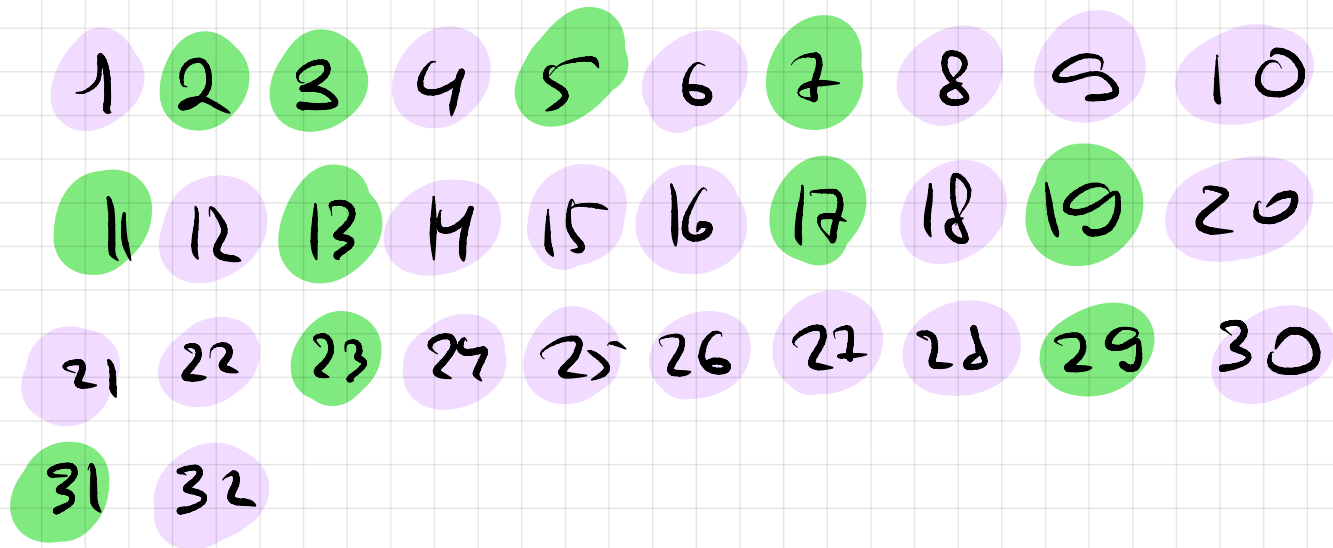
condition  $1 \leq \deg f < n$  with the

condition  $1 \leq \deg f \leq \lfloor \frac{n}{2} \rfloor$ . For,

if  $g$  is a product  $g = f \cdot h$  and  $\deg g = n$ ,

then  $f$  or  $h$  has the degree at most  $n/2$ .

One can also do something similar to the sieve of Eratosthenes



Can we reuse this idea for irreducible polynomials?

$p=2$  in  $\mathbb{F}_2[t]$

irreducible of degree one:  $t$ ,  $t+1$

irreducible of degree two:  $t^2+t+1$

irreducible of degree three:  $t^3+t+1$ ,  $t^3+t^2+1$

irreducible of degree four:  $t^4+t+1$ ,  $t^4+t^3+1$ ,  
 $t^4+t^3+t^2+t+1$

How do we test if  $g = t^8 + t^4 + t^3 + t + 1$  is irreducible? If  $g$  had a non-trivial factor, then it would have an irreducible factor of degree at most  $\leq 1 = \lfloor \frac{\deg g}{2} \rfloor$ . We've listed all these factors above. So, by calculating the remainders of division of  $g$  by these factors, we convince ourselves that  $g$  is irreducible.

**Remark** The (essentially) unique field with  $q = p^n$  elements is usually denoted by  $\mathbb{F}_q$  or  $\text{GF}(q)$ .

$\text{GF}$  abbreviates Galois field.

## 2.2 Inverting linear and affine maps

Let me recall that a map  $T: V \rightarrow W$  on vector spaces  $V$  and  $W$  over a field  $F$  is called  $F$ -linear if  $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$  holds for all  $\alpha, \beta \in F$  and all  $x, y \in V$ .

**Def** If  $T: V \rightarrow W$  is a linear map, then the map  $S: V \rightarrow W$  of the form  $S(x) = T(x) + u$ , where  $u \in W$ , is called affine.

We want to understand how to invert  
linear and affine maps  $F^n \rightarrow F^n$ .

**Proposition 29.** Let  $T: F^n \rightarrow F^m$  ( $m, n \in \mathbb{N}$ )

be an  $F$ -linear map, where  $F$  is a field.  
Then there exist values  $a_{ij} \in F$  ( $i=1, \dots, m$   
and  $j=1, \dots, n$ ) such that

for the input  $x = (x_j)_{j=1, \dots, n} \in F^n$

the output  $y = T(x)$  is the vector

$y = (y_i)_{i=1, \dots, m}$  is given by

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

Proof: Let's consider vectors  $e_1, \dots, e_n \in F^n$   
such that the  $i$ -th component of  $e_i$  is 1,  
whereas all the other components are 0,

$$e_1 = (1, 0, \dots, 0),$$

$$e_2 = (0, 1, 0, \dots, 0),$$

$\vdots$

$$e_n = (0, \dots, 0, 1).$$

We decompose  $x = (x_1, \dots, x_n) \in K^n$  as

$$x = x_1 e_1 + x_2 e_2 + \dots + x_n e_n. \text{ Then}$$

$$T(x) = T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n)$$

$$= x_1 \underbrace{T(e_1)}_{=: a_1} + x_2 \underbrace{T(e_2)}_{=: a_2} + \dots + x_n \underbrace{T(e_n)}_{=: a_n}$$

So,  $y_i = T(x)$  satisfies  $y = x_1 a_1 + \dots + x_n a_n$ .

Let  $a_j = (a_{ij})_{i=1, \dots, m} \in K^m$ . Then, for

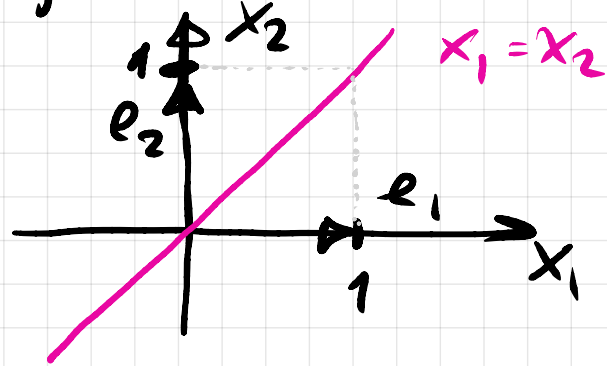
$y = (y_i)_{i=1, \dots, m}$ , we have

$$y_i = x_1 a_{i1} + x_2 a_{i2} + \dots + x_n a_{in}$$

$$= \sum_{j=1}^n a_{ij} x_j.$$

□

**Example** for  $F = \mathbb{R}$ , let's consider the orthogonal projection onto the line given by  $x_1 = x_2$ .



$T(x_1, x_2)$  is the orthogonal projection of  $(x_1, x_2)$  onto the line given by  $x_1 = x_2$ .

$$\begin{aligned} T(x_1, x_2) &= T(x) = T(x_1 e_1 + x_2 e_2) \\ &= x_1 T(e_1) + x_2 T(e_2) \end{aligned}$$

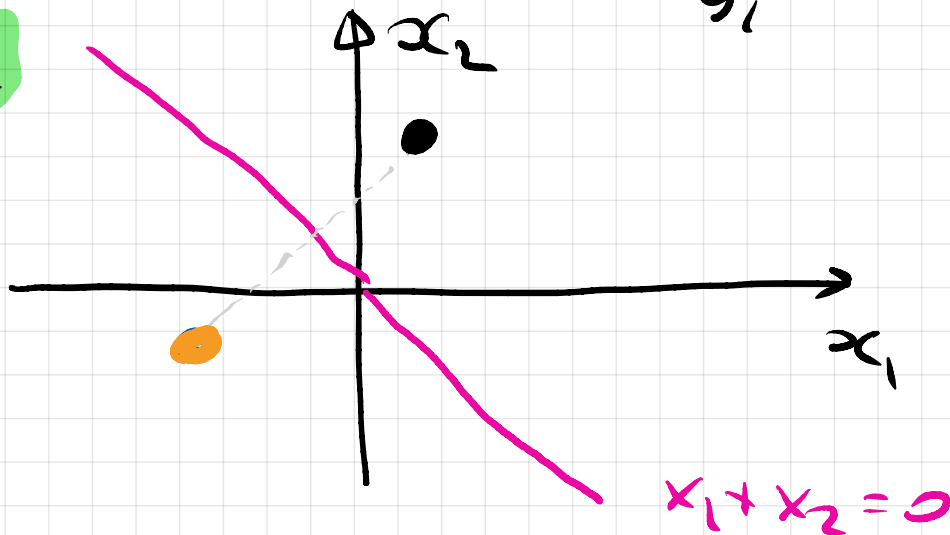
$$T(e_1) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$T(e_2) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$\Rightarrow T(x) = x_1 \cdot \left(\frac{1}{2}, \frac{1}{2}\right) + x_2 \cdot \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$= \left( \underbrace{\frac{1}{2}x_1 + \frac{1}{2}x_2}_{y_1}, \underbrace{\frac{1}{2}x_1 + \frac{1}{2}x_2}_{y_2} \right).$$

Example



Reflection  
with respect to  
the line given by  
the equation  $x_1 + x_2 = 0$   
(the field is  $\mathbb{R}$ ).

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$y = T(x) = x_1 T(e_1) + x_2 T(e_2) = x_1 \begin{pmatrix} 0 \\ -1 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

$$\left. \begin{array}{l} T(e_1) = -e_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \\ T(e_2) = -e_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \end{array} \right\} = \begin{pmatrix} -x_2 \\ -x_1 \end{pmatrix}$$

### Remark

A table of values  $a_{ij} \in F$  in a field  $F$  with  $i = 1, \dots, m$  and  $j = 1, \dots, n$  is called a matrix of size  $m \times n$  over  $F$ . The set of all such matrices ( $m \times n$  with entries in  $F$ ) is denoted as  $F^{m \times n}$ ;  $m$  is the number of rows and  $n$  is the

number of columns. For example,

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \in \mathbb{Q}^{2 \times 3}.$$

In linear algebra, one frequently prefers to interpret  $F^n$  as  $F^{n \times 1}$ , that is elements of  $F^n$  are viewed as columns.

So, instead of  $(2, 3, 4)$  one could also write  $\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ .

For a matrix  $A = (a_{ij})_{\substack{i=1, \dots, m \\ j=1, \dots, n}} \in F^{m \times n}$

and a vector  $x = (x_j)_{j=1, \dots, n} \in F^n$

one defines a matrix-vector multiplication

as  $y = A \cdot x$  with  $y = (y_i)_{i=1..m}$   
such that  $y_i = \sum_{j=1}^n a_{ij} x_j$ .

So, every linear transformation

$T: F^n \rightarrow F^m$  can be defined

by  $T(x) = A \cdot x$ , where  $A \in F^{m \times n}$ .

Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \cdot x_1 + 2 \cdot x_2 + 3 \cdot x_3 \\ 2 \cdot x_1 + 3 \cdot x_2 + 4 \cdot x_3 \end{bmatrix}$$



This matrix gives us a transformation  
from  $\mathbb{Q}^3$  to  $\mathbb{Q}^2$ .

**Definition** For two matrices

$$A = (a_{ij})_{\substack{i=1, \dots, m \\ j=1, \dots, n}} \in F^{m \times n} \quad \text{and}$$

$$B = (b_{jk})_{\substack{j=1, \dots, n \\ k=1, \dots, s}} \in F^{n \times s}$$

over a field  $F$  the product  $A \cdot B$  is defined as the matrix

$$C = (c_{ik})_{\substack{i=1, \dots, m \\ k=1, \dots, s}} \in K^{m \times s}$$

with

$$c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$$